

Frank Young

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Personal Summary

- Over 8 years of experience in heterogeneous platform (CPU/GPU/DSP/NPU) operator acceleration library, including DNN, BLAS, FFT, RAND. Deeply involved in 0-1 AI software stack development.
- 5 years of technical leadership experience, leading an engineering team at Biren Technology.
- Strong learning ability, pursuit of excellence, and a passion for technology.

Education

National Taiwan University |Telecom Institute| *Master's Degree* 2014.09—2017.01

- [1] **Yang, Hang**, D.-J. Deng, and K.-C. Chen, “On energy saving in ieee 802.11 ax,” *IEEE Access*, vol. 6, pp. 47 546–47 556, 2018.
- [2] **Yang, Hang**, D.-J. Deng, and K.-C. Chen, “Performance analysis of ieee 802.11 ax ul ofdma-based random access mechanism,” in *GLOBECOM 2017-2017 IEEE Global Communications Conference*, IEEE, 2017, pp. 1–6.

Chongqing University |Communication Engineering| *Bachelor's Degree* 2010.09—2014.06
Top 5% of the program, outstanding graduate.

Work Experience

Alibaba Damo |*Senior Software Engineer* 2024.12-Present

- Developing GEMM-like operators for next-generation AI inference chips (see Project Experience).

Biren Tech |*Senior Software Engineer/Manager* 2021.01—2024.11

- Built a BLAS acceleration library from scratch on a proprietary GPGPU platform, reaching A100-level throughput on first-generation chips (see Project Experience).
- Led the engineering team as Team Lead; owned technical roadmap and delivery.

Hikvision |*Software Engineer* 2017.07—2020.12

- Developed high-performance convolutional neural network (CNN) library for heterogeneous AI inference platforms, optimizing convolution and matrix operations on CPU/DSP/NPU.

Project Experience

CUTLASS Porting to New Chip |*Programming pattern* 2025.07—Present

- Studied the CUTLASS framework and CuTe layout algebra in depth, then designed architecture-specific kernel variants to port a generalized GEMM implementation to a new chip.
- Implemented generalized GEMM operators using CUTLASS programming patterns, combined with handwritten assembly for further performance optimization.

BLAS Library Dev |*Development and Maintenance from Scratch* 2023.06—2024.11

- Led the project, responsible for design, construction, development, testing, CICD, documentation, and management.
- Optimized performance across major LLM workloads (Qwen, Llama2, GPT series), reaching A100-level throughput on first-generation chips.
- Designed and implemented kernel selection algorithms.
- Implemented advanced optimization strategies: warp-specialized programming, fused bias/gradient accumulation, and BF16 accumulation.
- Implemented operator registration framework, logging system, and core operator development; filed 3 patents in AI acceleration.

Ultimate Optimization of GEMM |*based on verdi/zebu* 2022.09—2022.11

- Handwritten assembly to achieve ultimate performance GEMM on Vcore, achieved full utilization for specific shapes by observing waveforms through Verdi.
- Implemented simulated FP32 GEMM algorithm based on Tcore, achieved full utilization for specific shapes, and outperformed A100 in performance; filed 2 patents from this work.

- Developed a deep learning framework.
- Self-defined operator/graph expression based on YAML referencing ONNX.
- Compared precision with mainstream industry solutions and designed custom precision comparison solutions; registered a software copyright for the framework.

Miscellaneous

- **Technology Stack:** Performance optimization, Acceleration library dev, Deep learning model deployment(CNN, LLM).
- **Languages & Tools:** AI coding copilot, C++, Python, CUDA, Assembly, CMake, Verdi/SimVision, Git, JIRA, Jenkins.
- **IP:** 5 AI-domain patents, 1 communications patent, 1 software copyright.